

Relaxing the steady-state assumption displaces forest soil carbon models along the trade-off between litter input and turnover

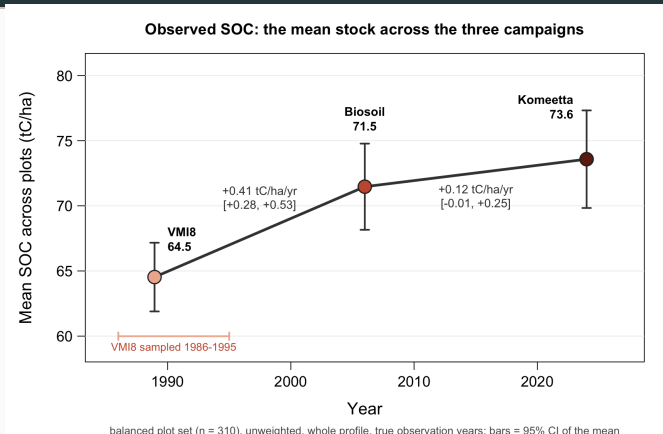
HIKET — internal progress talk: where the manuscript stands

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October 2026

Finnish forest soils have been measured three times



The stock rose by $+0.259 \pm 0.040 \text{ tC ha}^{-1} \text{ yr}^{-1}$ over the whole window, but only $+0.117 \pm 0.066$ over the last interval. The accumulation is slowing down.

Why the trajectory, and not the stock, is the quantity that matters

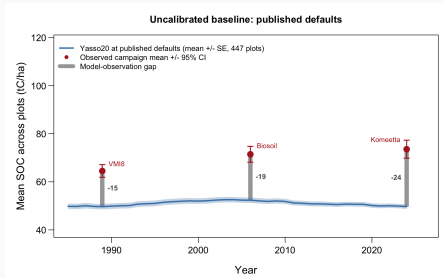
- A national inventory reports the **annual change** of the soil stock, not the stock.
- In Finland the soil term is **modelled**: NFI → litter → Yasso07 → Δ SOC. No soil measurement enters the chain.
- Forest-land mineral soils, 2022: sink of **4.8 Mt CO₂** against a biomass sink of 12.6 Mt; reported uncertainty **31.5%** vs 16.3%.
- The land-use sector is already a net source (+4.4 Mt CO₂e). A term of that size and uncertainty decides the sign.

The awkward step

A model has to be told how much C the soil holds on day one, and how it is split between pools — for a soil whose history was never observed.

Sweden measures its soil term by resampling its NFI plots (NID 2026). Most countries cannot afford that, and model it instead.

The convention: start the soil in balance with its litter input

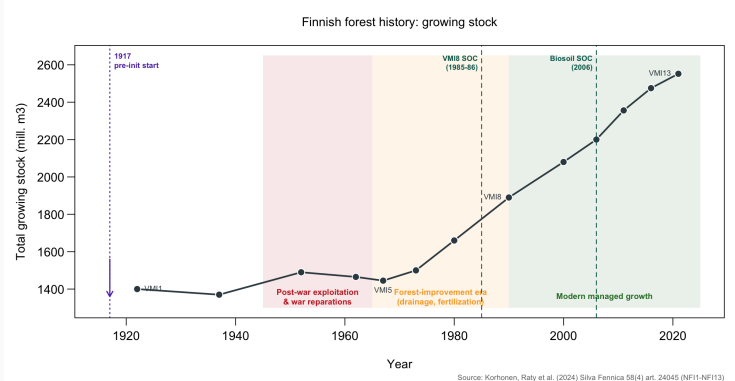


Yasso20, published parameters, steady-state start: flat and low.

- Shared across lineages: Yasso, RothC, Century, global land-model spin-ups; CBM-CFS3 cycles to a repeating average.
- The Finnish inventory spins up on NFI6 litter (1971–76) and 1960–90 climate; its defence is Peltoniemi et al. 2006 — a **precision** study around an assumed level (“precision, not accuracy”).
- Liski et al. 2006 is our direct antecedent: same country, model family and NFI litter — and they *assumed* the 1922 state, having no repeated soil measurement.

A soil started in balance has no accumulation left to perform.

Finland is the wrong place for that assumption



Growing stock +84% over a century, most of it after the late 1960s. Litter follows biomass, so the input the models must integrate has been rising the whole time — and unmanaged stands hold ~40% more mineral-soil C (Kumpu et al. 2025).

Three questions, weighted in this order

- Q1 Does a **transient, calibrated initial state** reproduce the observed national SOC trajectory — and what does reproducing it *require* of the models? (weight ~55%)
- Q2 Does it matter, for projection, **where a calibration lands** on the exchange between litter input and transit time? (weight ~25%)
- Q3 What part of the variation between **individual plots** can none of the models see? (weight ~20%)

Two quantities govern a stock

litter input J and mean transit time τ : at balance $C^* = J\tau$. A stock fixes the product only. A trajectory can do more.

Six models, two roles, one frame

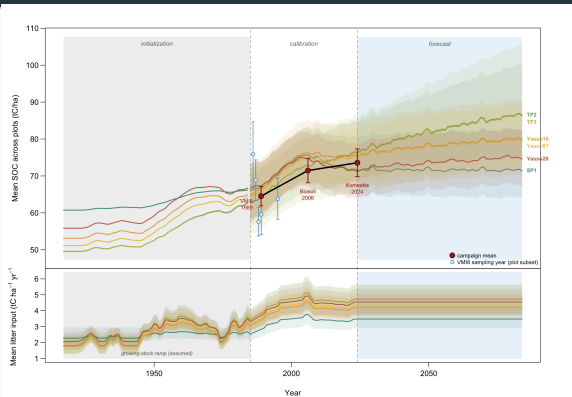
Model	Pools	Free	Role
SP1	1	6	benchmark
TP2	2	7	benchmark
TP3	3	9	benchmark
Yasso07	5	20	the inventory model
Yasso15	5	26	successor
Yasso20	5	26	successor

The simple models are a transparent ladder: any Yasso result is checked against a structure one can follow by hand.

Equal *external information*, not equal DoF

- Yasso: rates **fixed** at published values; the 12 transfer fractions, climate, woody size are free.
- SP1/TP2/TP3: rates **anchored on ICBM** (fast fixed, slow pinned by a tight prior); humification fractions free.
- Priors homogenised in *construction*, not in value: same rule per parameter class, centre from each model's own publication.
- All six integrate the linear system exactly; no explicit Euler step.

The move: estimate the starting state instead of assuming it



Start in 1917 in balance with the 1917 litter; raise the input along the Liski et al. 2006 reconstructed shape to 1985; then drive with the annual record. Two auxiliary parameters: $\sigma_{\text{init}} = J_{1917}/J_{1985}$ and σ_{input} , a multiplier on the whole litter series.

Data and error model, in one slide

Observations

- 456 NFI plots, 1205 plot-years; 365 fitted / 91 held out (by plot).
- Three campaigns rebuilt on **one basis**: bulk density, coarse fragments (median 43%, a $1.6\times$ effect), per-plot depth, modelled deep tail (validated on held-out 40–80 cm).
- Official 2006/2024 national stocks reproduced to 0.01 t/ha. First campaign: 2006 litter+moss layer added; **true sampling year** used.

Litter and climate

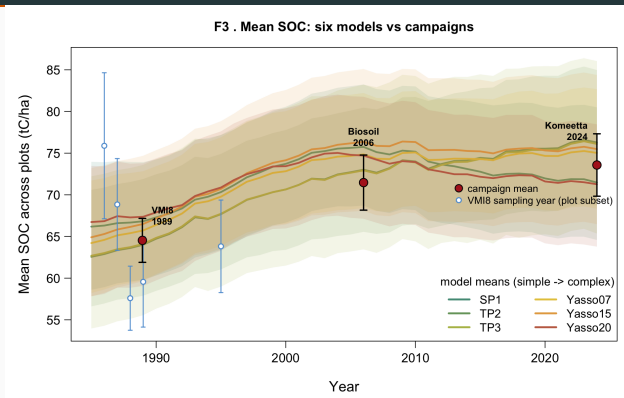
- Tupek et al. tree-litter product, $\text{AWEN} \times \text{size class}$; $\bar{J} = 2.511 \text{ tC ha}^{-1} \text{ yr}^{-1}$; understorey *not* included by design.
- FMI 10 km daily grid; climate modifier computed per year, then averaged (Jensen).

Likelihood

$$\log y_{ij} = \log f_{ij}(\theta) + u_{r(i)}^R + u_i^P + u_j^C + e_{ij}$$

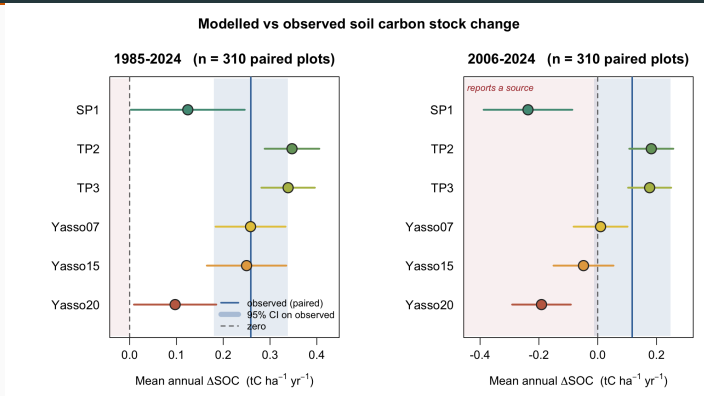
Total SD fixed at 0.800, split (never added): region 0.117, plot 0.396, campaign 0.06/0.03, residual 0.685. Offsets marginalised; Σ constant $\Rightarrow$ one triangular solve per evaluation.

All six models reach the observed stocks



No model sits more than 3.5 tC/ha from any campaign mean — with anchored rates, a constrained input and an estimated start. A precondition, not a result: any $J \times \tau$ with the same product reaches the same stock.

... and bracket the observed rate of change



Observed +0.259; Yasso07 +0.258, Yasso15 +0.250; TP2/TP3 stronger, SP1/Yasso20 weaker. The bracketing is the result. Which model lands closest is *not* interpretable — reported as consistency, never validation.

Model structure is nearly invisible in the fit

Model	Pools	Free	R^2 cal	R^2 hold	RMSE	Eff. flux
SP1	1	6	0.017	0.000	47	3.2
TP2	2	7	0.016	0.000	48	4.3
TP3	3	9	0.015	0.000	49	4.3
Yasso07	5	20	0.019	0.000	48	4.0
Yasso15	5	26	0.021	0.001	47	3.9
Yasso20	5	26	0.016	0.000	47	4.2

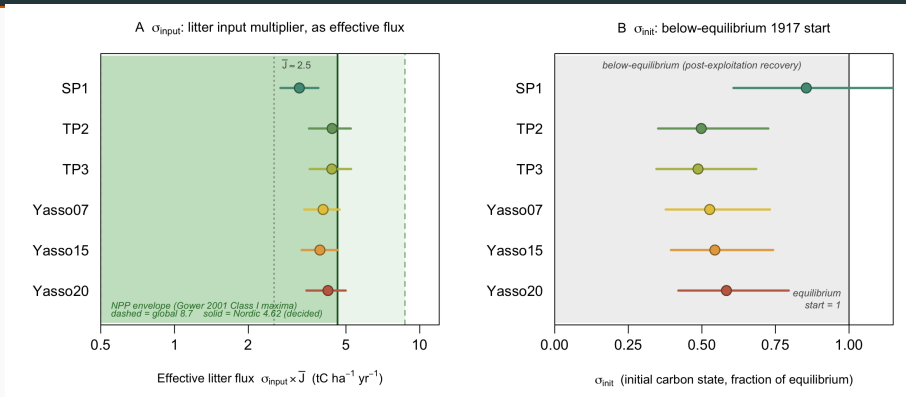
Table 1 of the draft. Effective flux = $\sigma_{\text{input}} \bar{J}$, $\text{tC ha}^{-1} \text{yr}^{-1}$; RMSE on held-out plots, tC/ha .

- Held-out R^2 is zero to three decimals in all six; RMSE spans 3.6% best to worst.
- Skill does not climb the pool ladder. The fit cannot choose a structure.
- Fractions converge ($R\text{-hat} \leq 1.008$) yet stay prior-pinned, anti-correlated within a pool, < 1 nat gained.

What three campaigns identify

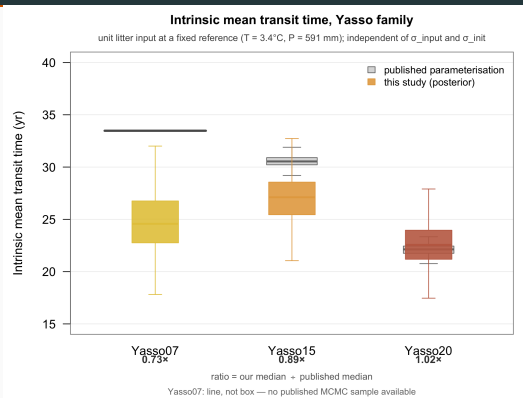
a stock and a rate of change — which is what the inventory needs, and all that is claimed.

What reproducing the trajectory requires: more litter input...



Left: effective flux $3.2\text{--}4.3 \text{ tC ha}^{-1} \text{ yr}^{-1}$ in all six, against the 2.5 the product supplies — inside the NPP envelope. Right: every model starts 1917 below its 1985 balance, the direction the history requires.

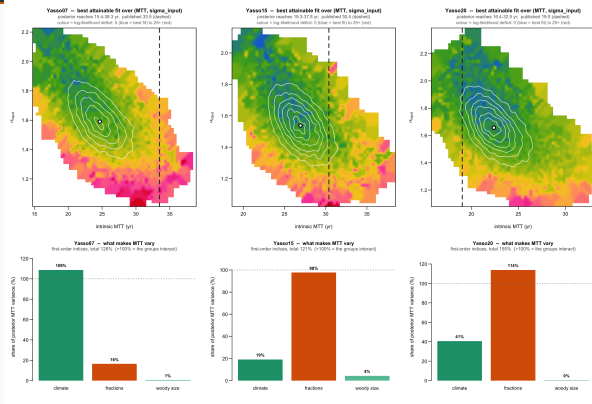
... and, for two of three Yassos, faster turnover



Intrinsic mean transit time at one reference climate and litter mix, from the equations alone.

Calibrated / published: 0.73 (Yasso07), 0.89 (Yasso15), 1.03 (Yasso20). **Yasso20 is the negative control** — a data-treatment artefact would have moved all three.

One displacement, not two: the trade-off



Colours = fit to the data alone; white contours = posterior (data + priors); dashed = published MTT. The trade-off exists under *either* start; the transient start narrows it, partially ($r = -0.64$ to -0.73).

The input displacement is set by the data, not by the prior

Two production runs, one factor

prior centre of σ_{input} : $1.08 \rightarrow 1.27$
(+18%)

posterior median	arm B (1.08)	arm A (1.27)
SP1	1.27	1.35
TP2	1.73	1.87
Yasso15	1.54	1.62

- The posterior moves only $\sim \frac{1}{3}$ as far as the centre.
- Under either anchor the posterior sits $1.2\text{--}1.6\times$ *above* its prior centre.
- So the requirement for more litter survives the choice of anchor; the anchor is a documentation question.

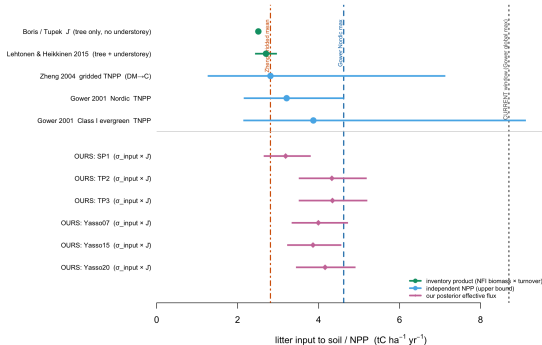
Decided: the paper stays on arm B (Lehtonen & Heikkinen understorey anchor); arm A is a sensitivity.

What might be missing from the litter input

Litter input to Finnish forest soil: inventory products vs independent NPP vs our calibration

Bars are reported ranges (Gower, Zheng) or 95% intervals (L&H, ours).

$\text{litter} = \text{NPP} - \text{biomass increment} - \text{stemwood removed} - \text{exudates}$.

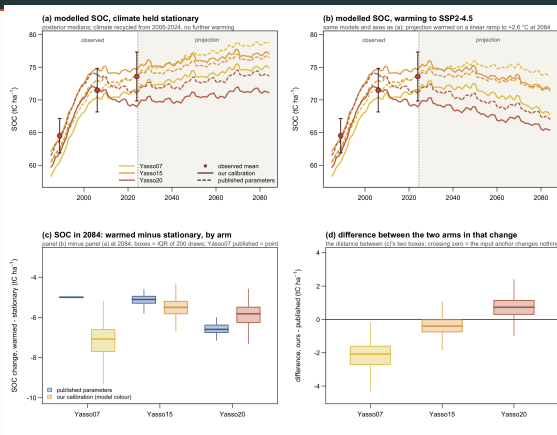


Gap to explain: $0.68\text{--}1.83 \text{ tC ha}^{-1} \text{ yr}^{-1}$

- **Understorey** — omitted by design; 0.51–0.67 in the NID itself.
- **Mycorrhizal residues** — not a biomass compartment; plausibly 0.2–0.5.
- **Rhizodeposition** — standard in cropland accounting (Bolinder 2007); absent from the forest chain.
- **Fine-root lifespan** — points the *other* way ($\sim 1/3$ of the turnover in use).
- **Early succession** — a timing gap after clear-cut.

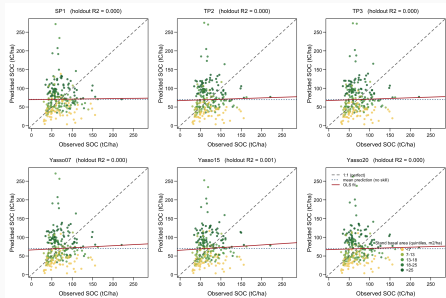
A set that brackets the gap, not a sum that matches it.

Published vs calibrated, run forward under the same warming



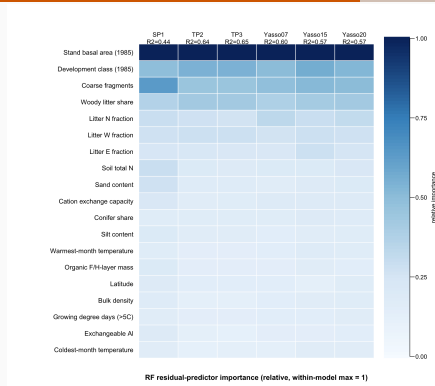
Neither arm recalibrated; only σ_{input} refitted so both reproduce the observed window. Only **Yasso07** separates (-2.1 tC/ha, $[-3.6, -0.8]$, SSP2-4.5). Yasso15/20 do not separate at all.

Between plots, the models describe a gradient the soil does not have



- Held-out $R^2 = 0$ in all six. What matters is what the models *predict*: stocks ordered by basal area, spread wider than observed.
- Slope of log stock on basal area: models $+0.044$ – 0.049 , observed $+0.0044$ (n.s. after region and soil).
- No model has a basal-area term: the **litter product** rises at $+0.072$ ($\times 18$ across the range) and the models pass it through.
- Woody-compartment mismatch tested and **refuted** (zeroing woody inputs: 6%; SP1/TP2/TP3 give the same slope with no composition at all).
- Plots cut in 1985–95 hold $\sim 40\%$ more C than predicted: the same missing flux, from the other end.

About half of what the models leave behind is recoverable



A random forest on the residuals recovers 44–65% out of bag, against an observation repeatability of ~ 0.49 . The same variables lead in all six: basal area, development class, coarse fragments. A productivity proxy and a soil property no model has a term for.

Conclusions

1. A transient, calibrated initial state reproduces a soil C dynamic a steady-state start **cannot** produce: the stocks, the rate, and an accumulation that slows down.
2. Reproducing it requires **more litter input** (all six) and, graded across the Yassos, faster turnover — *one* displacement along a trade-off the measurements narrow but do not resolve. It is set by the data, not the prior; the likeliest reading is an incomplete litter input.
3. The position on the trade-off reaches the **projection** for a model with one climate modifier (Yasso07) and not for models with pool-specific modifiers — a usable robustness statement for an inventory.
4. Headroom is real and finite: +28 to +49% to balance at present input (Liski 2006: +38%). Realising it costs decades of sustained input.
5. Half the plot-scale residual is recoverable from site variables that already exist — a well-posed next study.

Where the manuscript stands, and what is still open

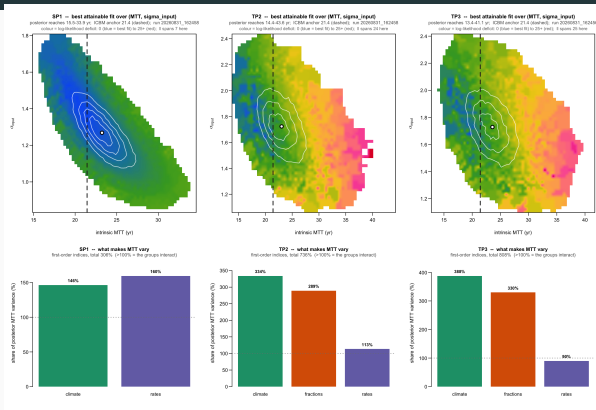
Done

- Draft v1, 36 pp, builds clean; Intro and M&M through two rounds of comments.
- Title chosen; mean *transit* time adopted (Sierra et al. 2017).
- Literature review, 8 topics; missing-flux budget.
- NextGenC deliverable rebuilt on the production run (maps, matrices).

Open

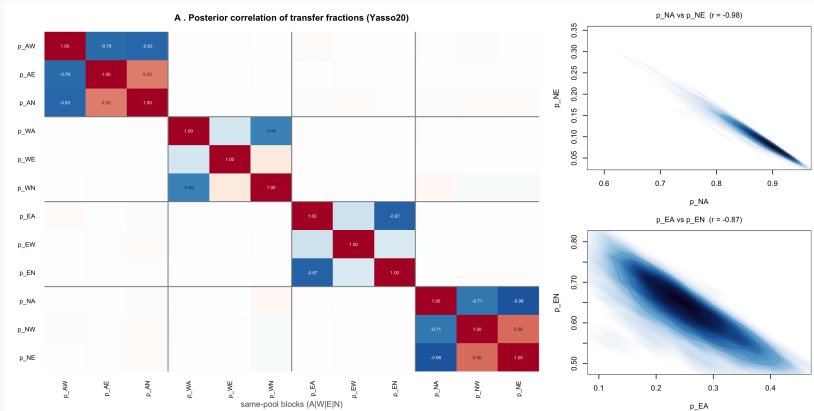
- **The equilibrium-init counterfactual has never been run** — the Introduction's central claim is still theory (design in `NEXT_RUN_equilibrium_init.md`).
- Abstract, Limitations, “other inventories” subsection, author block.
- Climate citation provisional; prior-criteria table unported.
- Data availability: the SOC compilation is not a citable source — settle with its holders.
- F14's Yasso20 dashed line still at the old 19.0 (corrected point 21.9): rebuild.

Backup — the benchmark trio rides the same ridge, more strongly



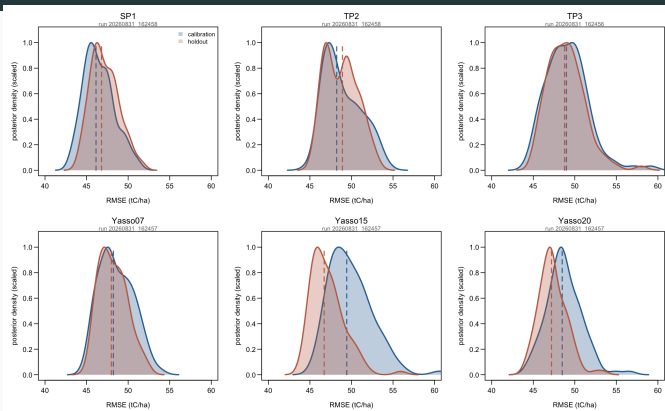
SP1 / TP2 / TP3, same construction as F14. With the climate modifier included the trio correlates -0.90 / -0.81 / -0.81 — stronger than any Yasso. The trade-off is a property of the data, not of the five-pool structure.

Backup — why the fractions are not identified (and why that is fine)



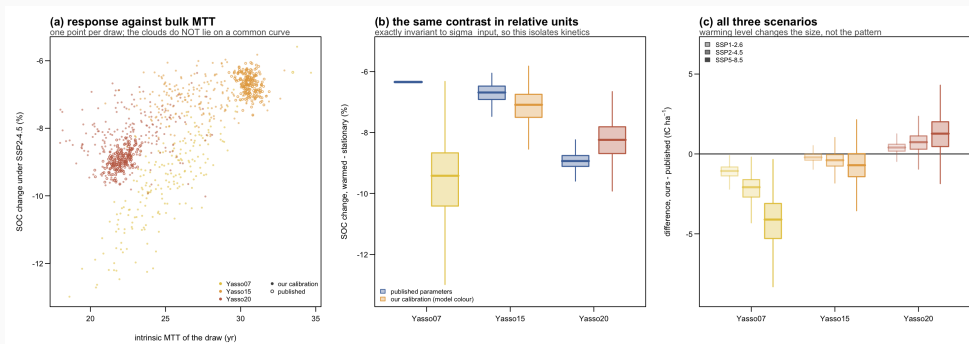
Fractions leaving the same pool are near-perfectly anti-correlated: the data fix the net flux out of a pool, not its split. Prior-pinned, ridge-correlated, < 1 nat — with clean R-hat.

Backup — how tightly is the fit pinned?



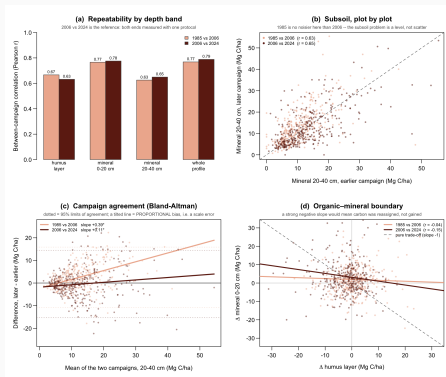
RMSE recomputed for every posterior draw. Location = fit; width = how much the posterior disagrees with itself. The cross-run comparison instrument, since log-likelihoods are not comparable across error models.

Backup — why only Yasso07 separates in projection



Yasso07 applies one ξ to every pool: $MTT = MTT_{ref}/\xi$ exactly, so turnover and temperature sensitivity are the same lever. Yasso15/20 carry ξ_{AWE} , ξ_N , ξ_H ; the humus pool that holds the carbon barely moves ($\times 1.04$ – 1.06).

Backup — the observed rate: what it rests on



- Basis (decided): balanced set $n = 310$, unweighted, whole profile, true years. Worth $1.8\times$ on the 2006–2024 rate (+0.117 vs +0.209 pairwise).
- 1985 organic layer was OFH only; the missing litter+moss layer (3.4–4.3 Mg C/ha) was 89% of the apparent 1985→2006 organic gain. Each plot's own 2006 LM added.
- “1985” is 1986–1995 (19.5% of plots in 1995): interval to 2024 is 35.1 yr, not 39.
- Full-window rate survives a 9% shared offset; the last-interval rate dissolves at 2%.

Backup — the error model, and why the total is above the measured spread

term	SD	measured on
region u^R	0.117	8 latitude bands
plot u^P	0.396	same-plot pair covariance
campaign u^C	0.060 / 0.030	prescribed
residual e	0.685	remainder
total	0.800	fixed

- 1205 plot-years are not independent: plot ICC 0.57–0.71 ($n_{\text{eff}} \approx 630$); latitude-band means vary 5–6 $\times$ more than iid allows.
- Uniform σ inflation broadens every direction by $\sqrt{k}$; only correlated structure broadens the *level* direction at fixed total variance.
- Design rule: residuals may *check* a design value, never set one.
- $\tau = 0$ reproduces the independent log-normal likelihood exactly (unit-tested).

Backup — the 1917 state the models imply

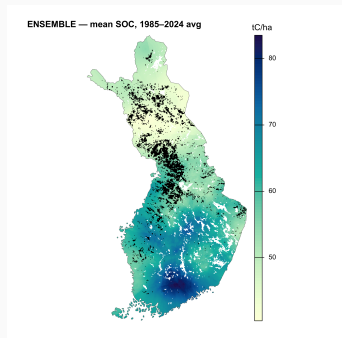
	σ_{init}	J_{1917}	J_{1985}	C_{1917}	C_{1985}	pre-run rate
SP1	0.63	2.32	3.68	39.5	49.1	+0.148
TP2	0.45	1.90	4.36	32.8	44.8	+0.181
TP3	0.43	1.73	4.14	32.6	44.6	+0.177
Yasso07	0.32	1.50	4.67	30.8	45.2	+0.217
Yasso15	0.29	1.27	4.22	30.1	47.0	+0.242
Yasso20	0.33	1.47	4.56	31.3	47.9	+0.235

Run 20260817 (the previous run; the current one moved σ_{init} up to 0.49–0.84). Fluxes in $\text{tC ha}^{-1} \text{yr}^{-1}$, stocks in tC/ha . The pre-run rate is *below* the observed +0.259 in all six. The 1917 stock (30–40 tC/ha) is **reported, not constrained**; the growing-stock bound on σ_{init} was retired because judging our start by an equilibrium benchmark re-imports the convention the paper argues against.

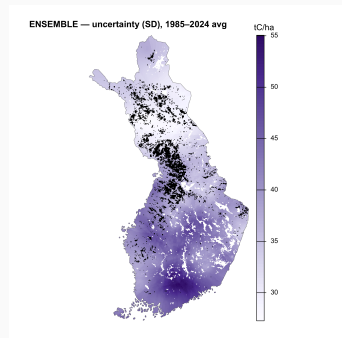
Backup — equilibrium headroom

- Hold litter at its present level and let each calibrated model run to its own balance: the soil sits **28–49% below** that balance in five of six models (SP1: +5%).
- Liski et al. 2006, from an assumed 1922 steady state and the same forward simulation: +38% “centuries later”.
- Two routes, one answer on *how far* the soil is from balance — and both say the room is finite.
- The main projection has **no climate change**: 2005–2024 climate recycled, litter frozen at 2024. The projected gain is relaxation towards balance, not a warming response.

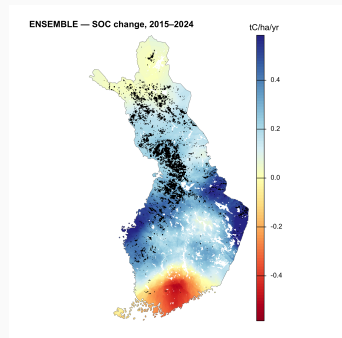
Backup — the deliverable: gridded SOC and its uncertainty



ensemble mean



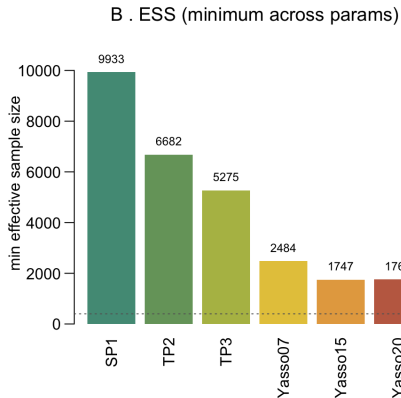
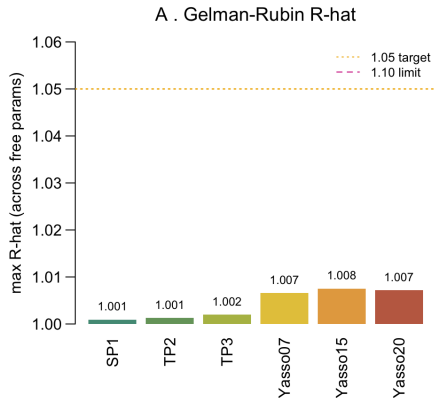
ensemble SD



decadal change

Six-model ensemble, kriged to 2 km, historical 1985–2024 only (no projection). NextGenC / Zenodo deposit, CC-BY-4.0. A bonus product, not a conclusion.

Backup — convergence



All parameters $R\text{-hat} \leq 1.008$, $ESS \geq 1450$, numerically impossible proposals $\sim 0\%$ in all six models (production run 20260831).